Myanmar Institute of Information Technology



I Semester 2025-2026

Special Project PROJ 4001 Mid-term Report

Smart Route & Info Guide for Bagan Pagodas

Presented by:

Phyu Hnin Aung (2022-MIIT-CSE-040)

Kyaw Zay Ya Hein (2022-MIIT-CSE-014)

Instructor in charge : Dr. Myat Thuzar Tun

Project Supervisor : Daw Khin Myat Nwe Win

Submitted in Partial Fulfilment of the Special Project

February 2, 2026

Abstract

Bagan, Myanmar, is renowned for its thousands of temples and pagodas spread across a large area, making navigation challenging for tourists. A prior web-based Tour Guide Assistant provided basic shortest-path routing but relied on continuous internet connectivity and lacked mobile usability and personalization. This project proposes a mobile-first, offline-capable solution titled "Smart Route & Info Guide for Bagan Pagodas," built with Flutter and Dart for cross-platform deployment on Android devices.

The system redesigns navigation architecture to support on-device route computation using the A* algorithm and cached OpenStreetMap tiles for reliable use in low-connectivity environments. New functional modules extend the system beyond navigation, including Hotels and Restaurants Discovery, Smart Itinerary Builder, Group Chat and Trip Rooms, Local Tips and Q/A Feed, and a Safety and Emergency Toolkit. A context-aware, bilingual AI chatbot provides location-based historical information and travel guidance. Firebase supports authentication, cloud synchronization, and user personalization when connectivity is available.

Overall, this project delivers a comprehensive, user-centric mobile solution that enhances navigation, cultural engagement, and travel safety in Bagan.

Contents

Abstract 1
Chapter 1 Introduction 3
1.1 Background 3
1.2 Objectives 3
1.3 Scope 4
1.4 Project Description 4
1.5 Project Schedule 4
Chapter 2 System Overview and Components 5
2.1 System Overview 5
2.2 Components 5
Chapter 3 System Design and Flowchart 6
3.1 System Design 6
3.2 Flowchart 6
Chapter 4 System Implementation 7
4.1 Implementation Steps 7
Chapter 5 Conclusion 8
5.1 Summary 8
5.2 Future Enhancements 8
References 9

Chapter 1
Introduction

1.1 Background

Bagan is one of Myanmar's most significant cultural heritage regions, containing over 2,000 ancient pagodas distributed across a wide archaeological landscape. Due to limited signage, large distances, and unreliable internet connectivity, tourists often face difficulties navigating between pagodas efficiently. Traditional guides and printed maps are not sufficient for optimized routing or real-time decision making. An earlier web-based Tour Guide Assistant provided shortest-path navigation but required continuous internet access and server-side processing, limiting practical use in Bagan.

This project addresses these limitations by designing an offline-first mobile architecture that supports on-device routing, personalized navigation, and contextual guidance. In addition, it introduces new modules for discovery, planning, social coordination, community knowledge, and safety, extending the scope from navigation to a full tourism companion.

1.2 Objectives

The objectives of this project are:

- To design and develop an offline-first mobile navigation system for cultural heritage tourism.
- To implement on-device shortest-path routing using the A* algorithm.
- To support personalized navigation through user profiles, visited-place tracking, and favorites.
- To integrate OpenStreetMap with offline tiles for reliable navigation in low-connectivity environments.
- To develop a context-aware AI chatbot that provides location-based historical information and travel guidance.
- To introduce new functional modules: discovery of nearby places, smart itinerary builder, group chat and trip rooms, local tips and Q/A feed, and a safety and emergency toolkit.
- To provide a bilingual user interface (English and Myanmar) for accessibility and inclusiveness.
- To evaluate system performance and usability in a mobile environment.

1.3 Scope

The project focuses on a mobile application for tourists in Bagan. The system supports offline navigation, pagoda information, and itinerary planning. It also supports optional online features such as authentication, cloud sync, and real-time group chat when internet is available. The initial deployment targets Android devices. Administrative tools for managing pagoda and service data are included. The system is designed to be extensible to other heritage sites.

1.4 Project Description

The Smart Route & Info Guide for Bagan Pagodas is a Flutter-based mobile application that provides intelligent, personalized, and socially enriched navigation. Tourists can view pagodas on an offline map, compute shortest paths using on-device A*, access cultural information, plan multi-stop itineraries, discover nearby amenities, and coordinate with travel groups. The application includes a bilingual, context-aware AI chatbot. Administrators manage data and moderate community content.

1.5 Project Schedule

Weeks 1-2: Requirement analysis, limitation review, and offline-first architecture design.
Weeks 3-4: UI/UX design, Firebase setup, and dataset preparation.
Weeks 5-6: Offline map integration and GPS tracking.
Weeks 7-8: On-device routing algorithm implementation.
Weeks 9-10: Personalization and advanced module development.
Weeks 11-12: AI chatbot integration and community/safety features.
Weeks 13-14: Evaluation, optimization, testing, and finalization.

Chapter 2
System Overview and Components

2.1 System Overview

The system is organized as a mobile client with offline-first capabilities and optional cloud synchronization. Core navigation functions operate without internet access using locally cached map tiles and a stored graph dataset. When connectivity is available, Firebase services enable user authentication, profile synchronization, and collaborative features such as group chat and shared itineraries.

2.2 Components

Client Application (Flutter + Dart):
- Map UI, route visualization, and navigation controls.
- Offline tile cache for OpenStreetMap.
- On-device A* routing engine with local graph storage.
- User profile, favorites, and visited-place tracking.
- Itinerary builder with multi-stop optimization.
- Discovery module for hotels, restaurants, and services.
- Group chat, trip rooms, and community feed.
- Safety toolkit and emergency features.
- Bilingual UI and AI chatbot integration.

Backend Services (Firebase):
- Authentication for user accounts.
- Firestore for user profiles, favorites, and chat content.
- Cloud storage for media and data synchronization.

Data Layer:
- Pagoda and POI dataset with metadata.
- Graph edges and nodes for routing.
- Local cache for offline discovery and tips.

Chapter 3
System Design and Flowchart

3.1 System Design

The architecture follows an offline-first model. The client loads map tiles and routing data from local storage. User requests for routes are computed on-device with A* using the stored graph. When internet connectivity is available, the system synchronizes user data with Firebase. Modules such as group chat and community feed use real-time updates but queue messages when offline.

Key design decisions include:
- Local routing engine to avoid dependency on external servers.
- Modular feature layout for discovery, itinerary, social, and safety modules.
- Bilingual content management with language toggling.
- AI chatbot bound to location context and user route state.

3.2 Flowchart

High-level user flow:
Start -> Open App -> Load Offline Map + Data
-> Select pagoda or POI -> Compute route (A*) -> Display route
-> Optional: Save favorites / Add to itinerary
-> Optional: Discover nearby places / Ask chatbot / Join group chat
-> End

Chapter 4
System Implementation

4.1 Implementation Steps

- Prepare pagoda dataset and construct graph nodes/edges.
- Integrate Flutter Map and configure offline tile caching.
- Implement A* routing service and route rendering with polylines.
- Add GPS tracking and user location marker.
- Build UI screens for discovery, itinerary, and favorites.
- Add Firebase authentication and Firestore data sync.
- Implement group chat, trip rooms, and community feed.
- Add safety toolkit features and emergency contacts.
- Integrate bilingual support and AI chatbot responses.
- Conduct offline testing and performance optimization.

Chapter 5
Conclusion

5.1 Summary

This project proposes a mobile-first, offline-capable tourism guide for Bagan that improves on an earlier web-based system. It delivers on-device routing, offline maps, personalized guidance, and a set of advanced modules for discovery, planning, social coordination, community tips, and safety. The architecture emphasizes reliability in low-connectivity environments while supporting cloud features when available.

5.2 Future Enhancements

Future work may include AR-based on-site guidance, deeper AI personalization, support for additional heritage sites, richer analytics for route suggestions, and expanded multilingual support.

References

[1] **OpenStreetMap**. https://www.openstreetmap.org
[2] **Flutter Framework**. https://flutter.dev
[3] **Firebase**. https://firebase.google.com